

tive mode. The simulator takes prerecorded videos from a forward-facing on-board camera connected to a human-driven data-collection vehicle, and generates images that approximate what would appear if the CNN were instead steering the vehicle. These test videos are time-synchronised with the recorded steering commands generated by the human driver. Since human drivers don't drive in the centre of the lane all the time, there is a need to manually calibrate the lane's centre as it is associated with each frame in the video used by the simulator.

The simulator transforms the original images to account for departures from the ground truth. Note that this transformation also includes any discrepancy between the human driven path and the ground truth. The transformation is accomplished by the same methods. The simulator accesses the recorded test video along with the synchronised steering commands that occurred when the video was captured. The simulator sends

the first frame of the chosen test video, adjusted for any departures from the ground truth, to the input of the trained CNN, which then returns a steering command for that frame.

The CNN steering commands as well as the recorded human-driver commands are fed into the dynamic model of the vehicle to update the position and orientation of the simulated vehicle. In Fig. 4, the green area on the left is unknown because of the viewpoint transformation. The highlighted wide rectangle below the horizon is the area which is sent to the CNN.

The simulator then modifies the next frame in the test video so that the image appears as if the vehicle was at the position that resulted by following steering commands from the CNN. This new image is then fed to the CNN and the process repeats. The simulator records the off-centre distance (distance from the car to the lane centre), the yaw, and the distance travelled by the virtual car. When the off-centre distance exceeds

one metre, a virtual human intervention is triggered, and the virtual vehicle position and orientation is reset to match the ground truth of the corresponding frame of the original test video.

Fig. 5 shows how the CNN learns to detect useful road features on its own, with only the human steering angle as training signal. It was not explicitly trained to detect road outlines. CNNs are able to learn the entire task of lane and road following without manual decomposition into road or lane marking detection, semantic abstraction, path planning and control. A small amount of training data from less than a hundred hours of driving is sufficient to train the car to operate in diverse conditions, on highways, local and residential roads in sunny, cloudy and rainy conditions.

Image identification in autonomous vehicles

To be able to identify images, autonomous vehicles need to process a full 360-degree dynamic environment. This creates the need for dual-frame processing because collected frames must be combined and considered in context with each other. A vehicle can be equipped with a rotating camera to collect all relevant driving data. The machine must be able to recognise metric, symbolic and conceptual knowledge as demonstrated in Fig. 6.

Metric knowledge is the identification of the geometry of static and dynamic objects, which is required to keep the vehicle in its lane and at a safe distance from other vehicles. Symbolic knowledge allows the vehicle to classify lanes and conform to basic rules of the road. Conceptual knowledge allows the vehicle to understand relationships between traffic participants and anticipate the evolution of the driving scene. Conceptual knowledge is the most important aspect for being able to detect specific objects and avoid collisions.

One current method of obstacle detection in autonomous vehicles is



Fig. 4: Simulator in interactive mode

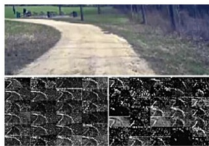


Fig. 5: How the CNN sees an unpaved road. Top: Camera image sent to the CNN; bottom left: activation of the first-layer feature maps; bottom right: activation of the second-layer feature maps

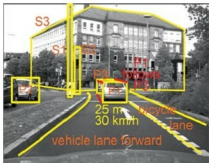


Fig. 6: Image identification in autonomous vehicles